

Embedded System

- An embedded system is a computing system with a dedicated function within a larger mechanical or electrical system, often with real-time constraints.
- 98% of all microprocessors are manufactured components of embedded systems.

Where do you find Computers?



ES Application Domains

Home & Building Automation

- Bringing intelligence, convenience and lifestyle



Smart Energy

- Adding power awareness to products and helping to save energy



Multimedia

- Wireless audio streaming and advanced remote controls



Security and Safety

- Improving remote control and home monitoring



Industrial M2M Communication

- Internet enhanced M2M communication using existing Wi-Fi infrastructure



Embedded System Market

- Average household uses about 225 embedded processors; about 35 for the automobile itself
- Estimated 5 billion embedded processor in use --- 94% share of world market; Intel Pentium, Motorola PowerPC, etc. combined is only 6%

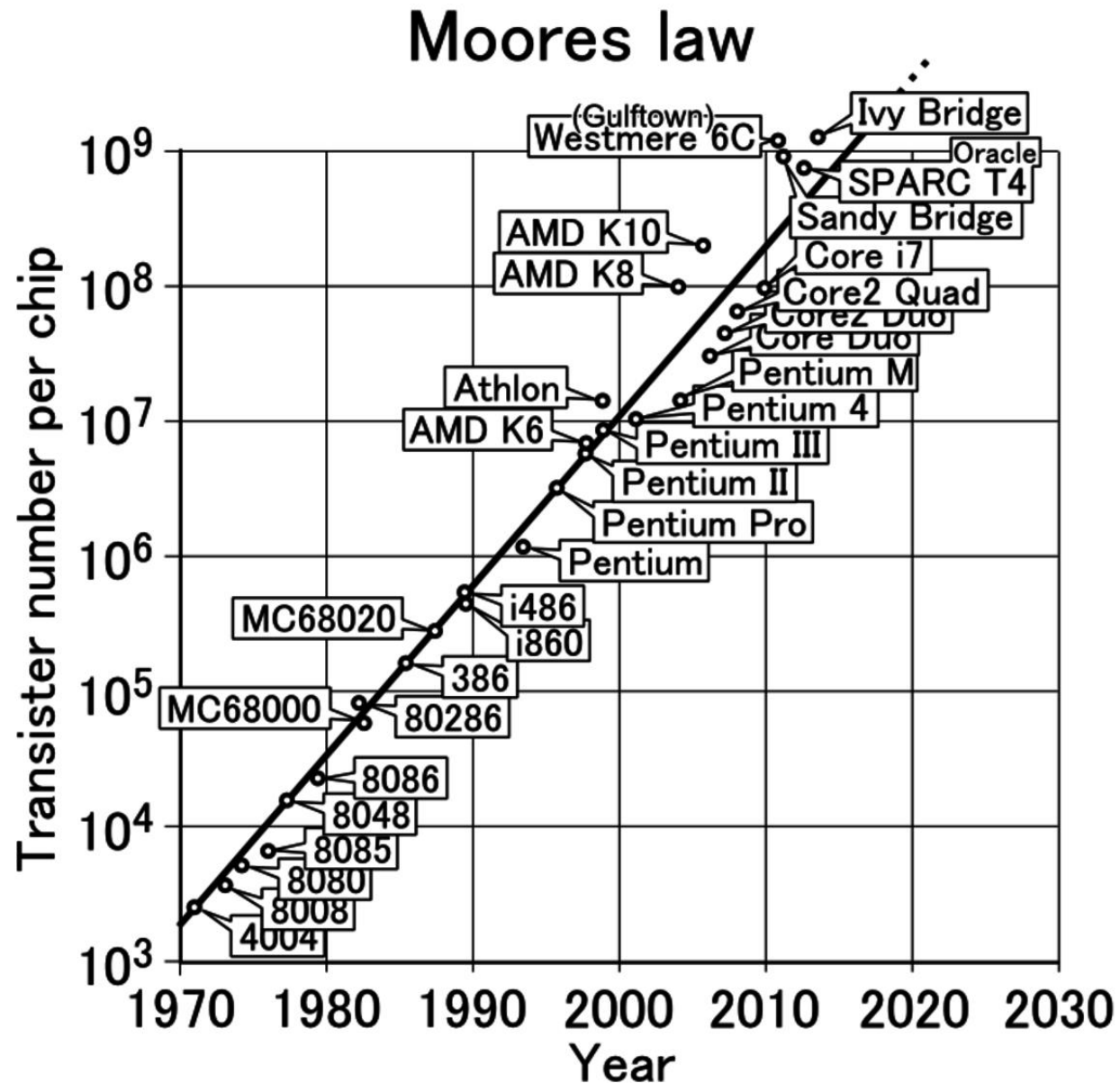
Why Embedded Systems

- Key elements of our society today
- Many new applications every day
- Requires knowledge of both software & Hardware
- Experienced embedded systems designers are in high demand and short supply

ES Modern Trends

- More and more Software processors and custom processors – thanks to IC technology
- More high-level languages in embedded domain – thanks to better compilers
- More and more synthesis tools in embedded domain – thanks to better synthesis tools

Technology Trend



Microcontrollers

x86

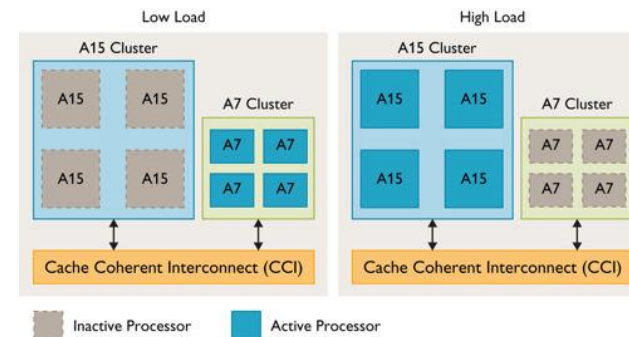
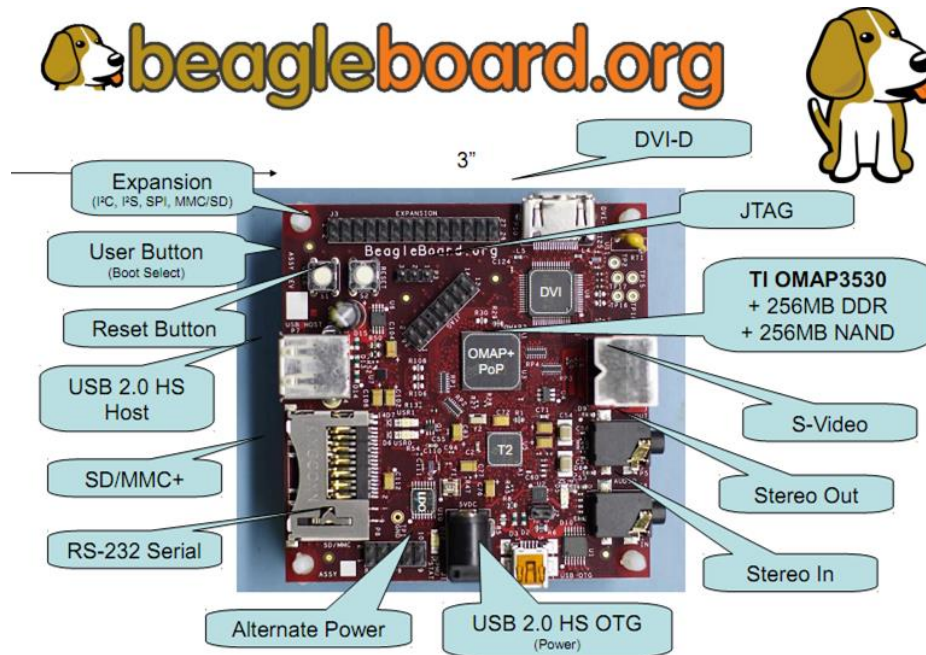
PIC

ARM

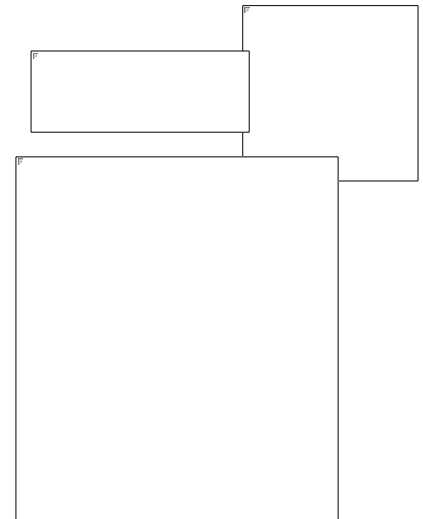
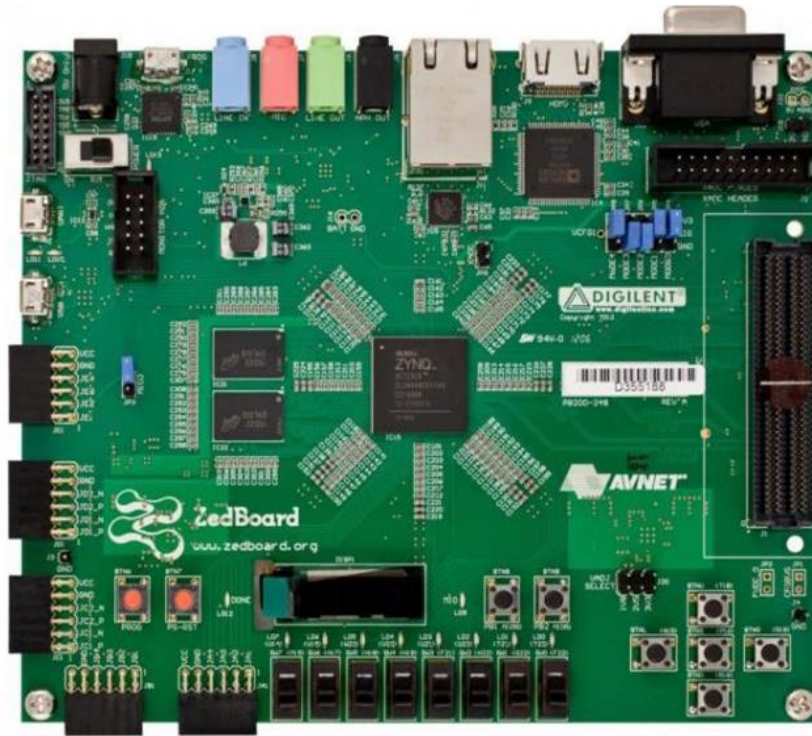
E8

AVR

Application Processor



Custom Processor



Course Objectives

- Introduce the trends and challenges in the design of embedded systems and teach chip technologies and design tools needed for these systems.

Course Organization

- 16 weeks (2+1 hr theory)
- 3 assignments
- 6 quizzes
- 2 sessional exams
- 1 final exam

Course Time Table

- Tue 0920 – 1100 hrs [Lecturer]
- Thu 08:30 – 11:00 hrs [Lab]

Course Grading



Course Books

- Embedded System Design: A Unified Hardware/Software Approach by Frank Vahid and Tony Givargis
- The 8051 Microcontroller, 4th Edition, I. Scott MacKenzie and Raphael C.-W. Phan
- The 8051 Microcontroller and Embedded Systems, 2nd Edition by Muhammad Ali Mazidi and others.
- [online]
https://www.tutorialspoint.com/embedded_systems/es_microcontroller.htm

Reading for Next Week

- Section 1.3: Embedded processor technology
- Section 1.4: IC technology